Correction: AI-driven Framework for Automated Competency Formalization: From Professional Standards to Adaptive Learning Outcomes

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Abstract

The rapid evolution of the labor market necessitates innovative approaches to align higher education curricula with professional standards. This study presents an AI-driven framework utilizing the GPT model to automate the formalization of professional competencies and learning outcomes from unstructured textual sources, such as professional standards and job descriptions. By transforming unstructured industry standards and job descriptions into structured competency maps, the framework ensures alignment with labor market needs. These maps are integrated into learning management systems (LMS) such as Canvas and Moodle, enabling the development of adaptive curricula. The methodology was validated using a dataset of professional standards from various industries, achieving a 30% increase in semantic accuracy compared to traditional methods. In addition, a multi-class classification task using Multinomial Naive Bayes, Gaussian Naive Bayes, and Random Forest models classified learning outcomes across college, undergraduate, graduate, and doctoral levels, achieving an accuracy score of 0.98, further confirming their applicability across qualification systems. Challenges such as

technological inequalities and lack of pedagogical flexibility remain. This scalable approach enables educational institutions to bridge the gap between academia and industry, helping to produce employable graduates.